Minimum-Phase and All-Pass Systems

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Key Concepts

1) A system with all of its poles and zeros located inside the unit circle is called a minimum-phase system.

2) A system with frequency response magnitude equal to one for all frequency is called an all-pass system.

3) The poles and zeros of an all-pass system must occur in complex conjugate reciprocal pairs. A pole at radius r and angle ϕ is paired with a zero at radius r^{-1} and angle ϕ.

4) Any system can be written as the product of minimum-phase and all-pass systems

    a) Zeros outside the unit circle are put into the all-pass system.

    b) Poles are added to the all-pass system in conjugate reciprocal locations to the zeros.

    c) Zeros are added to the minimum-phase system at the same locations as the poles that are added to the all-pass system.

5) The minimum-phase portion of any system has a stable and causal inverse.

6) The all-pass portion of a system does not have a stable and causal inverse.

7) Distortion in the magnitude response can always be inverted with a stable and causal system, unless there is a zero on the unit circle.

8) Distortion in the phase response may not be invertible with a stable and causal system.

Video Screenshots

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A stable, causal system has a stable, causal inverse if and only if all poles and zeros are inside $|z|=1$

called: **Minimum phase system**

Can show that phase lag of a system with poles/zeros inside $|z|=1$ is less than that of any other system with identical magnitude response

Any rational system function $H(z)$

$$H(z) = \underbrace{H_{min}(z)}_{\text{minimum phase}} \underbrace{H_{ap}(z)}_{\text{all pass}}$$

no zeros on $|z|=1$

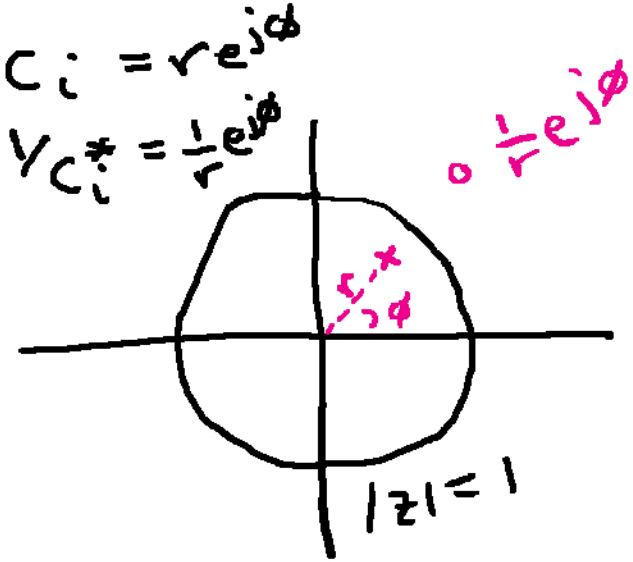
All pass: $|H_{ap}(e^{j\omega})| = 1$

All pass $\Leftrightarrow$ poles and zeros in conjugate reciprocal pairs

$$H_{ap}(z) = \prod_{i=1}^{P} \frac{z^{-1} - c_i^*}{1 - c_i z^{-1}}$$

poles: $c_i = re^{j\phi}$

zeros: $1/c_i^* = \frac{1}{r}e^{j\phi}$



To show $|H_{ap}(e^{j\omega})| = 1$, consider $P = 1$

$|e^{-j\omega}| = 1$

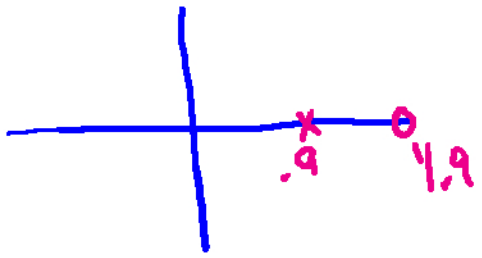
$$|H_{ap}(e^{j\omega})| = \left| \frac{e^{-j\omega} - c^*}{1 - c\,e^{-j\omega}} \right| = \left| \frac{e^{-j\omega}(1 - c^* e^{j\omega})}{1 - c\,e^{-j\omega}} \right|$$

$$= \left| \frac{1 - c^* e^{j\omega}}{\underbrace{1 - c\,e^{-j\omega}}_{b}} \right| = \frac{|b^*|}{|b|} = 1$$

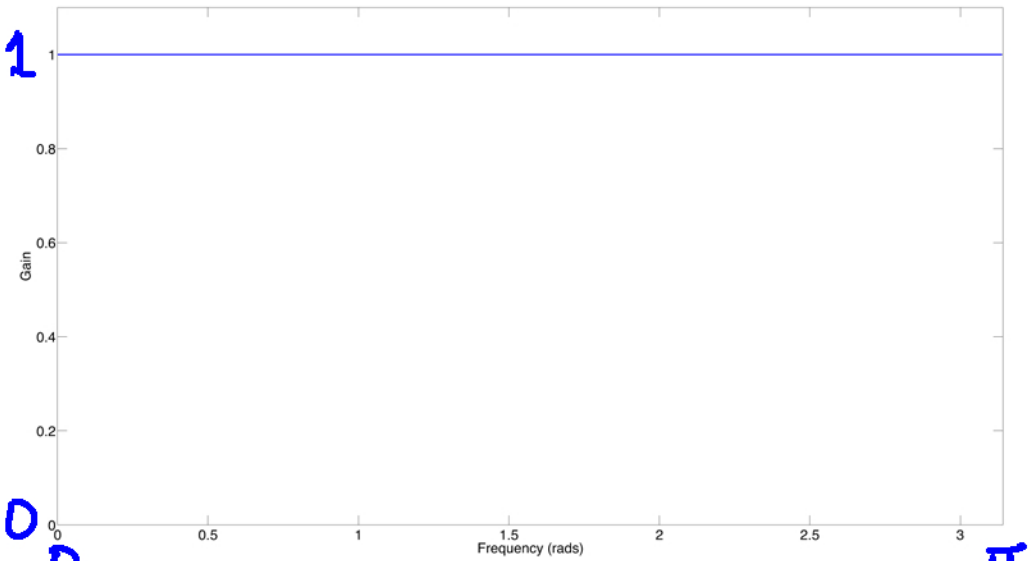
$$\frac{z^{-1} - c^*}{1 - c\,z^{-1}}$$

Example: All-pass System

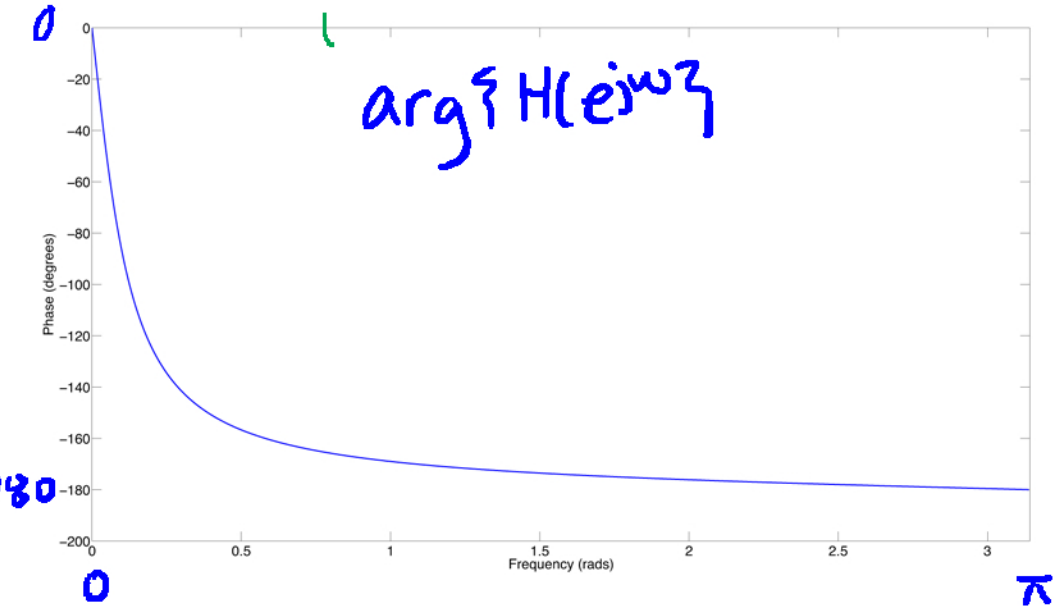
$$H(z) = \frac{z^{-1} - c^*}{1 - c\,z^{-1}}$$



$|H(e^{j\omega})|$

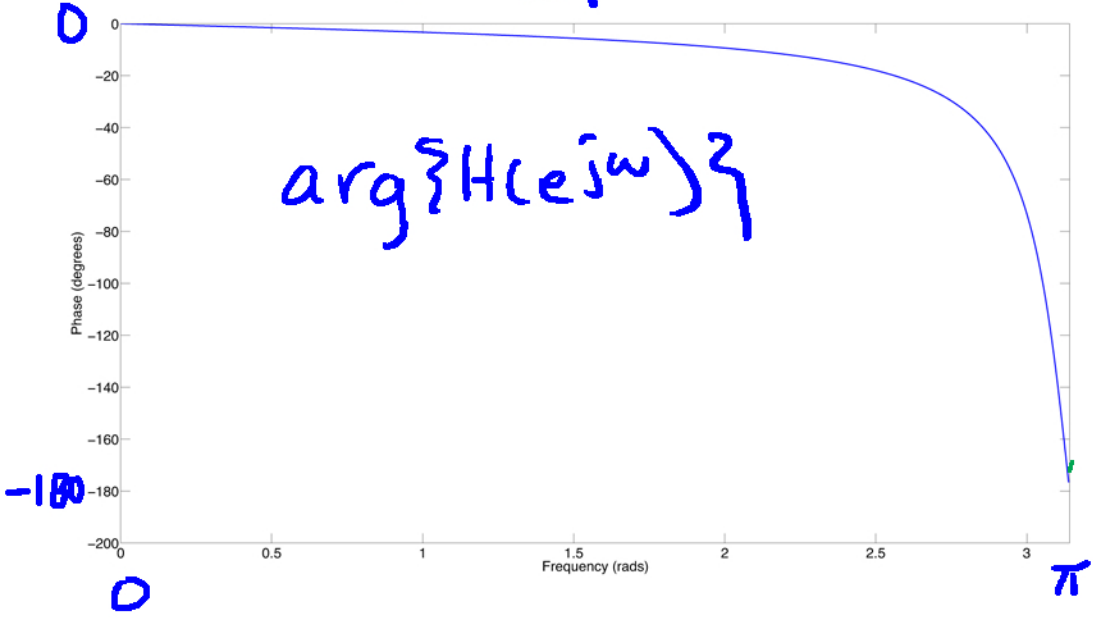


$c = 0.9$



$\arg\{H(e^{j\omega})\}$

$c = -0.9$



$\arg\{H(e^{j\omega})\}$

To factor $H(z) = H_{min}(z)\, H_{ap}(z)$:

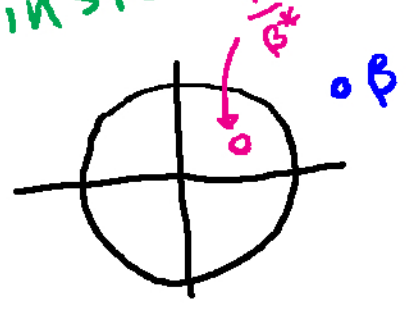
1) Take zeros that lie outside $|z|=1$ and move to $H_{ap}(z)$
2) Add poles to $H_{ap}(z)$ in conjugate reciprocal locations of zeros
3) Put zeros in $H_{min}(z)$ to cancel poles added to $H_{ap}(z)$

Suppose $H(z) = H_1(z)(1-\beta z^{-1})$; $|\beta|>1$, $H_1(z)$ min phase

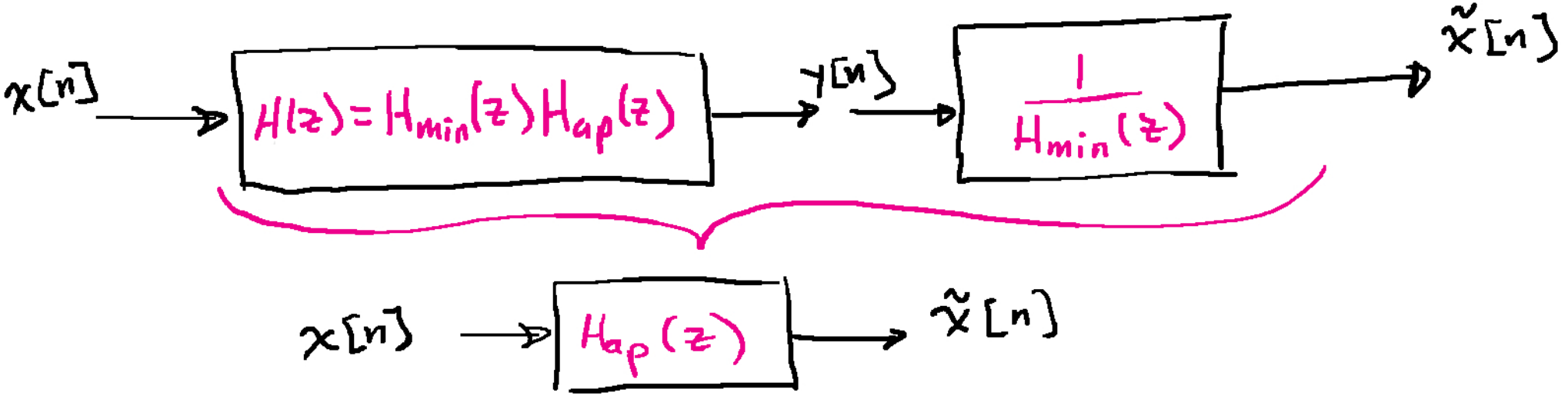
1) $H(z) = H_1(z)(-\beta)\left(z^{-1} - \frac{1}{\beta}\right)$

2+3) $H(z) = \underbrace{H_1(z)(-\beta)\left(1 - \frac{1}{\beta^*} z^{-1}\right)}_{H_{min}(z)} \cdot \underbrace{\frac{z^{-1} - \frac{1}{\beta}}{1 - \frac{1}{\beta^*} z^{-1}}}_{H_{ap}(z)}$

zero at $z=\beta$ in $H(z)$ reflected inside $|z|=1$



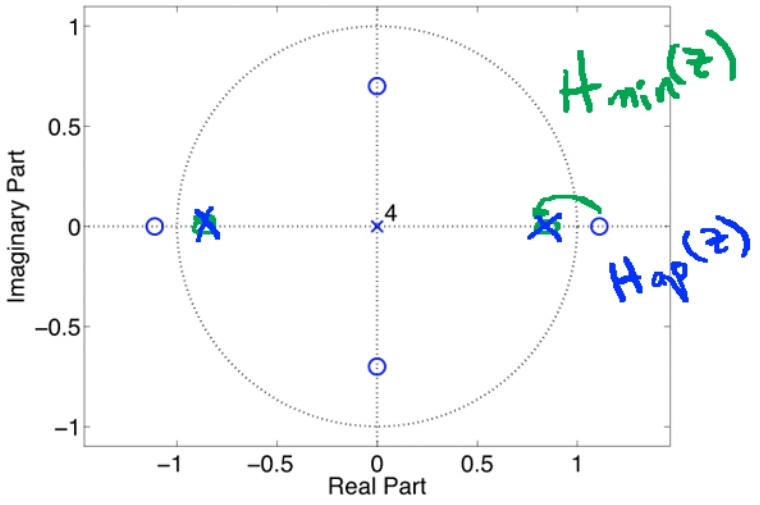
The minimum phase portion of any system has a stable, causal inverse system



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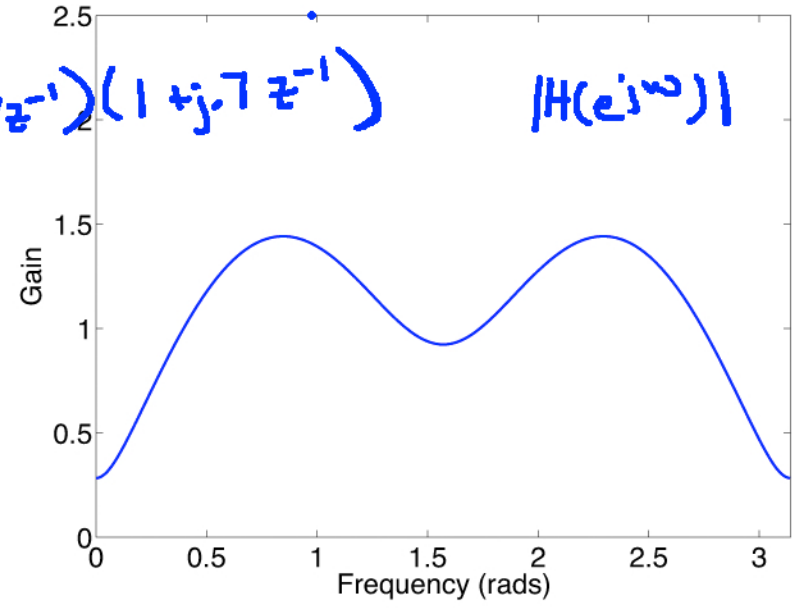
Example:

$$H(z) = \left(1 - \tfrac{1}{.9}z^{-1}\right)\left(1 + \tfrac{1}{.9}z^{-1}\right)\left(1 - j.7z^{-1}\right)\left(1 + j.7z^{-1}\right)$$

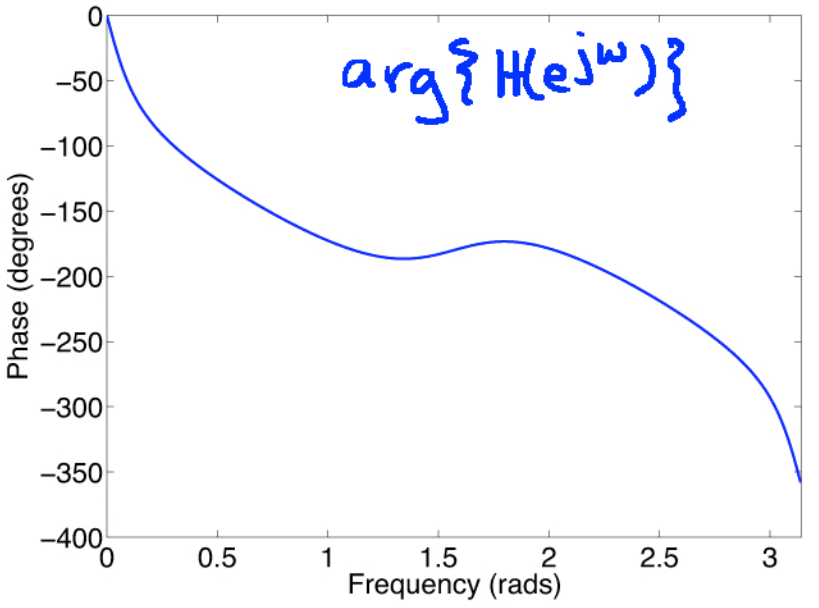


$H_{min}(z)$

$H_{ap}(z)$



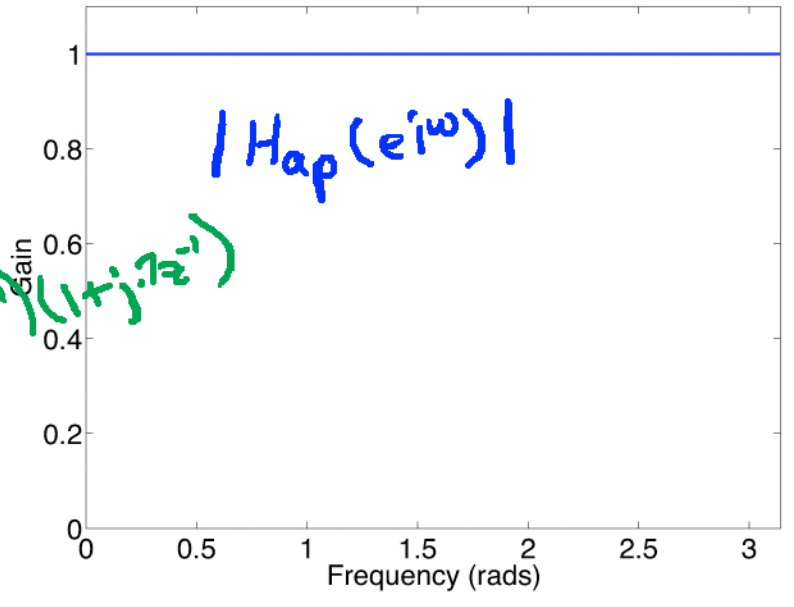
$|H(e^{j\omega})|$



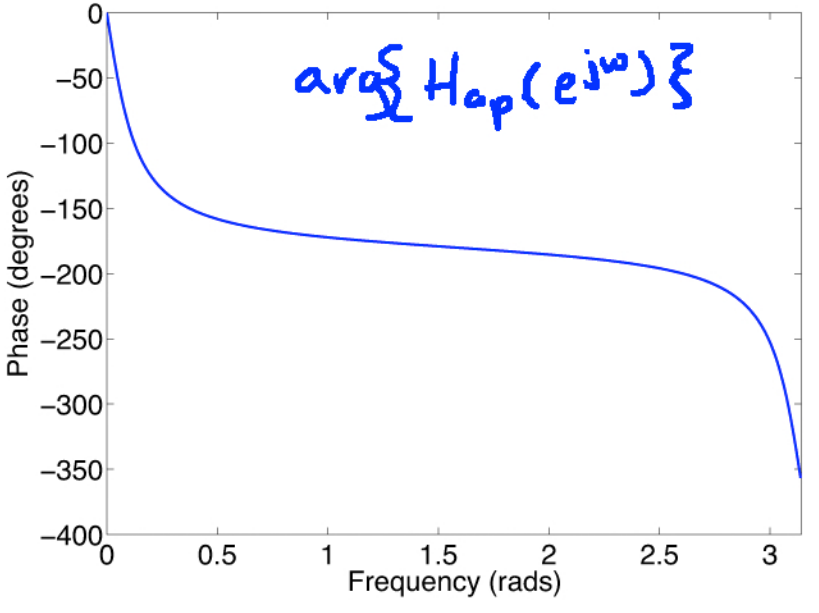
$\arg\{H(e^{j\omega})\}$

$$H_{ap}(z) = \frac{(z^{-1} - .9)(z^{-1} + .9)}{(1 - .9z^{-1})(1 + .9z^{-1})}$$

$$H_{min}(z) = \frac{1}{.81}(1 - .9z^{-1})(1 + .9z^{-1})(1 - j.7z^{-1})(1 + j.7z^{-1})$$



$|H_{ap}(e^{j\omega})| = 1$



$\arg\{H_{ap}(e^{j\omega})\}$

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